Snabel Al-Mashan Contracting Co.

Electrical Contractor • Saudi Electricity Company (SEC) Approved
HSE & Environmental Affairs Division

INTEGRATED ENVIRONMENTAL REGISTRY

System Management Documentation

Periodic Environmental Management System — ISO 14001:2015

ISO 14001:2015 **Saudi EMS Compliant** **GHS / SDS** **OHS-PR-02-09**

Document Preparer & Administration

Prepared By:	**Salman K. Alwajaan**	**Document No.:**	**ENV-2026-001**
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1 Document Control Card

1.1 Document Control Card

Field	Details	Field	Details
Document Title	Integrated Periodic Environmental Registry	Document No.	ENV-2026-001
Department	HSE & Environmental Affairs Division	Revision	Rev. 03
Issue Date	17 May 2026	Review Date	1 January 2027
Scope	All Projects + Chemical Warehouse	Total Pages	36 pages
Document Preparer	Salman K. Alwajaan	Role	Environmental Officer
Project Manager	____________________	Signature	________________
HSE Manager	____________________	Signature	________________
Approved by (CEO)	____________________	Signature	________________

1.2 Document Revision History

Rev.	Date	Description of Change	Prepared by	Approved by
01	May 2026	Initial Issue — Full document created	Salman K. Alwajaan	
02	June 2024	Chemical Warehouse Section Added	Salman K. Alwajaan	
03	June 2025	ISO 14001 Full Alignment + ERA Added	Salman K. Alwajaan	
04	__________	____________________	____________________	

1.3 Document Distribution & Storage

#	Copy Holder	Department	Copy Type	Date Issued	Signature
1	CEO / Executive Director	Executive Office	Controlled		
2	HSE Manager	HSE Department	Controlled		
3	Environmental Officer	HSE Department	Master (Original)		
4	Project Manager	Projects Department	Controlled		
5	Warehouse Supervisor	Warehouse	Controlled		

Storage: Master copy: SharePoint / HSE Drive / ENV-Docs / ENV-2026-001
Physical copy: HSE Manager Office, File Cabinet No. 3. **Retention:** 5 years minimum.

Contents

2 Glossary of Terms & Abbreviations

2.1 Environmental Management Abbreviations

Abbrev.	Full Term	Abbrev.	Full Term
EMS	Environmental Management System	**EHS**	Environment, Health and Safety
HSE	Health, Safety & Environment	**QHSE**	Quality, Health, Safety & Environment
EIA	Environmental Impact Assessment	**EMP**	Environmental Management Programme
HIRA	Hazard Identification and Risk Assessment	**CAPA**	Corrective and Preventive Actions
NCR	Non-Conformance Report	**KPI**	Key Performance Indicator
ERA	Environmental Risk Assessment	**SDS**	Safety Data Sheet
PPE	Personal Protective Equipment	**RACI**	Responsible, Accountable, Consulted, Informed
WEEE	Waste Electrical & Electronic Equipment	**ERP**	Emergency Response Plan
GHS	Globally Harmonized System	**WBGT**	Wet Bulb Globe Temperature
TWA	Time Weighted Average	**LOTO**	Lockout/Tagout

2.2 Air Quality & Pollution Parameters

Symbol	Parameter	Symbol	Parameter
PM2.5	Particulate Matter $\leq 2.5\ \mu m$	**PM10**	Particulate Matter $\leq 10\ \mu m$
NOx	Nitrogen Oxides	**SOx**	Sulfur Oxides
CO	Carbon Monoxide	**CO_2**	Carbon Dioxide
VOC	Volatile Organic Compounds	**GHG**	Greenhouse Gases
PCB	Polychlorinated Biphenyls	**dB(A)**	A-weighted Decibels
EMF	Electromagnetic Field	**LV/MV/HV**	Low/Medium/High Voltage
GIS	Gas Insulated Switchgear	**UPS**	Uninterruptible Power Supply
BAT	Best Available Techniques	**Scope 1/2/3**	GHG Emissions Categories

3 Environmental Policy

Official Environmental Policy — ISO 14001:2015

Snabel Al-Mashan Contracting Co. is committed to conducting its electrical contracting operations in an environmentally responsible manner, ensuring the protection of the environment, prevention of pollution, and reduction of adverse environmental impacts from all company activities executed for Saudi Electricity Company (SEC).

The company commits to full compliance with all applicable environmental regulations in the Kingdom of Saudi Arabia, including the Environment Regulation (Royal Decree No. M/165 of 2021), hazardous waste management regulations, SEC contractor requirements (OHS-PR-02-09), and the ISO 14001:2015 Environmental Management System standard.

We are dedicated to continual improvement through effective natural resource management, waste reduction and recycling, safe chemical storage and handling, and promotion of environmental awareness among all employees and subcontractors.

Senior management commits to providing all resources necessary for implementing, maintaining, and reviewing this EMS.

ENVIRONMENTAL OFFICER	HSE MANAGER	PROJECT MANAGER
______________________	______________________	______________________
Date:__________	Date:__________	Date:__________

CEO Commitment & Approval

I, the undersigned CEO, hereby approve and endorse this Integrated Environmental Registry.

CEO / EXECUTIVE DIRECTOR

____

Name:______________

Date:______________

Stamp:

4 Company Information

4.1 Company Identity & Contact Details

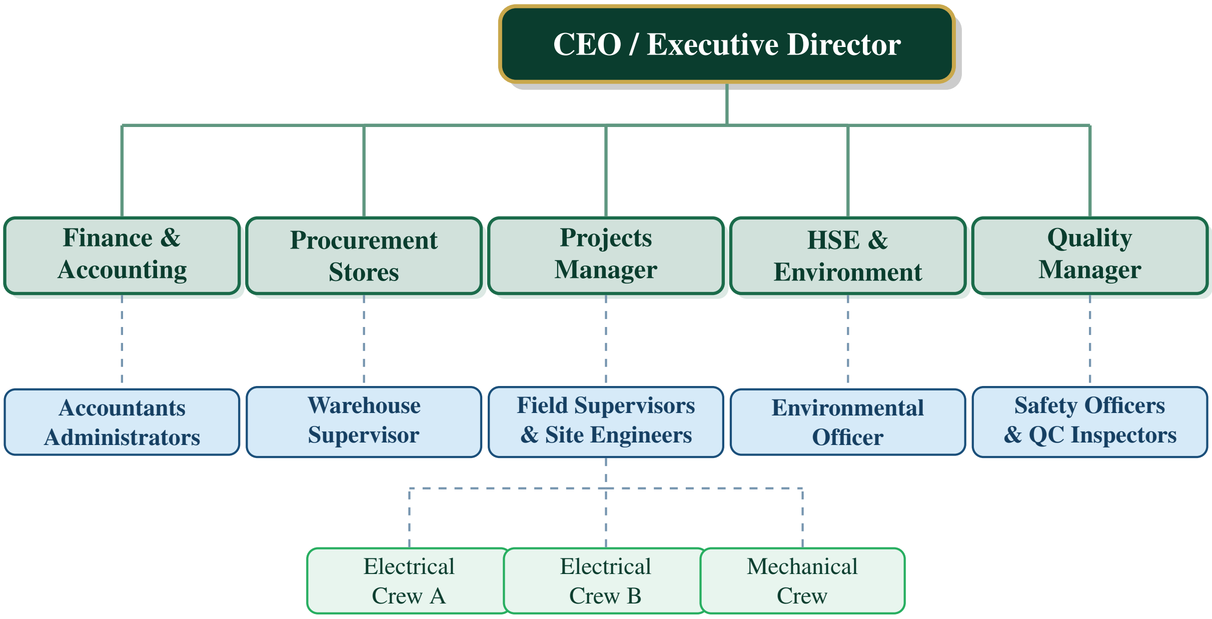
Field	Details
Company Name (Arabic)	____
Company Name (English)	Snabel Al-Mashan Contracting Co.
Commercial Registration No.	____
Facility Coordinates (GPS)	Lat: _______________ Long: _______________
Head Office Address	____
Official Email	____
Telephone / Fax	____
Total Workforce	_______________ employees
Work Hours per Day	__________ hours Shifts: ________
Shift 1 / Shift 2 Hours	_______________ / _______________
SEC Vendor No.	____
Environmental License No.	____

4.2 Facility Satellite Image

Figure 4.1: Facility Satellite Image

5 Organizational Structure

5.1 Company Organizational Chart



5.2 Responsibilities Matrix

Department / Role	Responsible Person	Environmental Responsibilities
CEO / Executive Director	______________	Approve environmental policy; provide resources; final regulatory authority
HSE & Environmental Manager	______________	Oversee EMS implementation; manage audits; regulatory liaison
Environmental Officer	Salman K. Alwajaan	Prepare registry; conduct inspections; compile KPIs; update ERA
Projects Manager	______________	Ensure site compliance; approve site environmental plans
Field Supervisors	______________	Daily environmental checks; waste segregation; incident reporting
Warehouse Supervisor	______________	Chemical storage management; SDS maintenance; Spill Kit readiness

5.3 RACI Matrix — Environmental Responsibilities

RACI Key: R = Responsible (does the work) **A** = Accountable (owns the outcome) **C** = Consulted (provides input) **I** = Informed (kept updated)

Activity	CEO	HSE Mgr	Env. Officer	Project Mgr	Field Supv.	WH Supv.	Safety Off.	Quality Mgr
EMS Policy Approval	A	C	C	I	I	I	I	I
ERA Preparation	I	A	R	C	C	C	C	I
Chemical Warehouse Mgmt	I	A	C	I	I	R	I	I
Hazardous Waste Disposal	I	A	R	C	R	R	I	I
Environmental Inspections	I	C	A/R	C	R	R	R	I
Spill Response	I	A	C	C	R	R	R	I
Environmental Training	I	A	R	C	R	R	R	I
KPI Monitoring & Reporting	I	A	R	C	C	C	C	C
Regulatory Compliance	A	R	R	C	C	I	I	I
ISO 14001 Audit	A	R	R	C	C	C	C	R
Incident Investigation	I	A	R	C	R	R	R	C
SDS / Chemical Register Update	I	A	R	I	I	R	I	I

5.4 Certifications & Accreditations

Certificate	Cert. No.	Issue Date	Expiry Date	Target
ISO 14001:2015 EMS	___________	___________	___________	Maintain certification
ISO 45001 OHS	___________	___________	___________	Zero LTI target
SEC Contractor License	___________	___________	___________	Renew 30 days prior
Environmental Storage License	___________	___________	___________	Annual renewal
NCEC Registration	___________	___________	___________	Active registration
MWAN Registration	___________	___________	___________	Licensed disposal

6 Scope of Work & Projects

Project Type	Description of Works	Key Environmental Aspects
LV/MV Electrical Distribution	Cable installation, panels, transformers, overhead lines	E-waste, oil spills, dust, noise
HV Substation Works	Erection at 33kV/132kV, switchgear, busbars	SF6 gas, PCBs, transformer oils
Intelligent Control Systems	HPS/LED lighting, BMS, SCADA	E-waste, chemical cleaners
UPS & Battery Systems	UPS installation, battery banks	Lead-acid batteries, acid risk
O&M Services	Preventive/corrective maintenance, industrial safety	Used oils, waste fluids, PPE
Underground Cable Laying	Trenching, ducting, reinstatement	Dust, soil disturbance, noise
Indoor Fit-out Works	MCC panels, metering, control systems	Construction waste, solvents
Multi-Energy Projects	Solar PV, battery storage integration	E-waste, chemical batteries
Telecom & Metering	Smart meters, fibre optics, BAS	E-waste, minimal chemical use
HV Overhead Lines	380kV/220kV/11kV line works	EMF, height risk, SF6

7 Organizational Context Analysis

7.1 Internal & External Context

Type	Context Factor	Description & Relevance	Impact Level
Internal	Organisational Structure	Hierarchical; clear HSE reporting line from site to CEO	High
Internal	Chemical Warehouse	On-site hazardous materials storage creates direct environmental liability	Very High
Internal	Fleet & Equipment	Diesel fleet generates direct emissions; maintenance generates waste oils	High
Internal	Management Commitment	CEO-endorsed policy; EMS budget approved annually	High
External	Saudi Environment Regulation	Royal Decree M/165 (2021) — mandatory compliance	Very High
External	SEC Requirements	OHS-PR-02-09 — direct contractual and environmental obligations	Very High
External	Regulatory Agencies	NCEC, MWAN, Civil Defense — inspection authority and licensing body	Very High
External	Vision 2030	Saudi National Transformation; drives demand for clean energy projects	High
External	Climate & Weather	Extreme heat (>45°C), sandstorms affecting operations	Medium

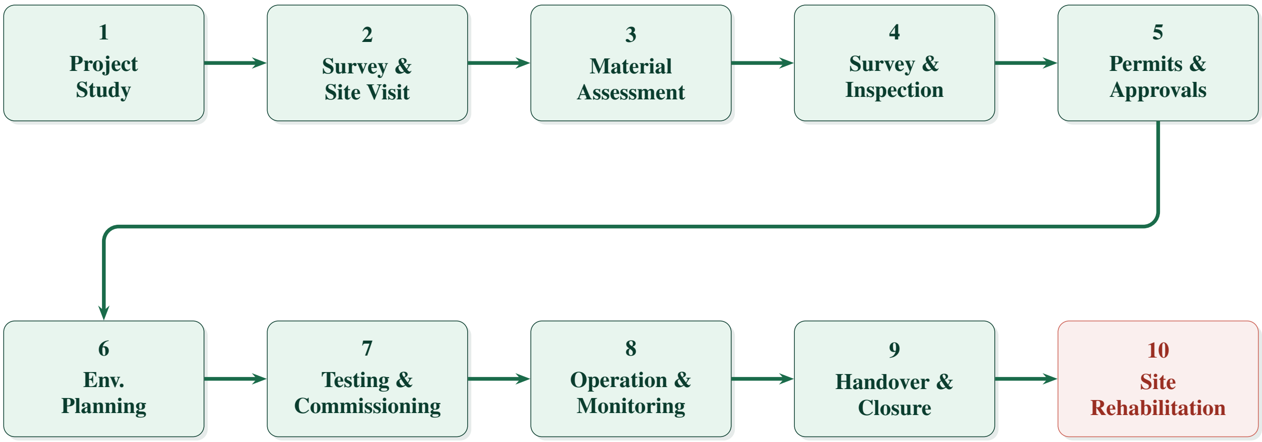
7.2 Stakeholder Analysis

Stakeholder	Influence	Environmental Expectation	Communication	Priority
Saudi Electricity Co. (SEC)	Direct/High	Full compliance with OHS-PR-02-09; zero incidents	Reports + Audit	**Very High**
Ministry of Environment	Regulatory	Compliance with Regulation M/165	Official reports	**Very High**
Civil Defense Authority	Regulatory	Safe chemical storage; approved ERP; annual inspection	Inspections	**Very High**
MWAN	Regulatory	Licensed waste disposal; manifests; zero illegal dumping	Manifests	**High**
Employees & Contractors	Internal	Safe pollution-free work environment	Training	**High**
Local Community	External	No dust, noise, spills affecting daily life	Community meetings	**Medium**

7.3 SWOT Analysis

STRENGTHS	WEAKNESSES
• ISO 14001:2015 certified EMS in place • Dedicated Environmental Officer • CEO-endorsed environmental policy • GHS-compliant chemical warehouse • Established SEC contractor relationship • Documented waste management procedures	• Multiple dispersed work sites hard to monitor • Heavy reliance on subcontractors • Variable environmental awareness among workers • Limited on-site emission monitoring equipment • No formal carbon footprint baseline established
OPPORTUNITIES	**THREATS**
• SEC sustainability strategy + green procurement • Saudi Vision 2030 driving clean energy projects • Circular economy incentives for waste recycling • Growing demand for ISO 14001 certified contractors • Carbon credit market development in KSA	• Tightening NCEC and Civil Defense enforcement • Chemical spill / warehouse fire risk • Regulatory penalties for non-compliance • Extreme heat and sandstorms (PM10, WBGT risk) • Rising cost of licensed waste disposal

8 Project Lifecycle Map



#	Phase	Key Environmental Impacts	Controls Required
1	Project Study	Minimal — desk-based study	Document environmental baseline
2	Survey & Site Visit	Noise, vehicle emissions, minor disturbance	Pre-wetting, low-emission vehicles
3	Material Assessment	Chemical identification; waste classification	SDS review; waste planning
4	Survey & Inspection	Dust, soil disturbance, access track creation	Silt fencing; haul road demarcation
5	Permits & Approvals	None direct	Obtain environmental permits; notify NCEC
6	Environmental Planning	Planning phase only	Complete ERA; prepare EMP; brief all staff
7	Testing & Commissioning	Oil leaks, SF6, battery acid, noise	Spill Kit ready; SF6 recovery; PPE
8	Operation & Monitoring	Ongoing emissions, maintenance wastes	KPI tracking; quarterly monitoring
9	Handover & Closure	Waste accumulation; site clearance	Manifests; soil testing; photographic record
10	Site Rehabilitation	Soil restoration; topsoil reinstatement	Geotechnical sign-off; re-vegetation

9 Equipment Fleet & Environmental Impacts

#	Equipment Type	Qty	Fuel Type	CO_2 Level	Noise dB(A)	Waste Generated
1	Hydraulic Excavator (Medium)	2	Diesel	High	85–95	Used oil, filters
2	Grader (Motor Grader)	1	Diesel	High	80–90	Used oil, filters
3	Bulldozer	1	Diesel	High	82–92	Used oil, filters
4	Forklift (Counterbalance)	2	Diesel/LPG	Medium	75–85	Used oil, batteries
5	Heavy Transport Trucks	4	Diesel	High	80–88	Used oil, tyres
6	Diesel Generators	3	Diesel	Medium-High	70–82	Used oil, fuel waste
7	Light Vehicles & Utilities	6	Petrol/Diesel	Low-Medium	65–75	Used oil, tyres
8	Welding Machines	4	Electric/Diesel	Low	70–80	Welding rods, fumes
9	Oxygen/Gas Cutting Sets	3	LPG/O_2	Low	65–75	Gas cylinders
10	Electrical Power Tools	10+	Electric	Negligible	70–85	E-waste, cables
11	Bobcat / Skid Steer	1	Diesel	Low	78–88	Used oil, filters
12	Water Tankers	2	Diesel	Low-Medium	68–78	Waste water

Equipment Relative CO_2 Emission Index

Equipment	CO_2 Level	Equipment	CO_2 Level
Excavator	85/100 High	Light Vehicles	45/100 Low-Med
Grader	80/100 High	Welding Machines	20/100 Low
Bulldozer	82/100 High	Gas Cutting Sets	15/100 Low
Forklift	50/100 Medium	Power Tools	5/100 Negligible
Transport Trucks	88/100 High	Bobcat	55/100 Medium
Diesel Generators	65/100 Med-High	Water Tankers	40/100 Low-Med

10 Transformer Management & Maintenance

10.1 Transformer Inventory

Reference Standards: IEC 60076 (Power Transformers) | IEC 60296 (Transformer Oil) | IEC 62271 (HV Switchgear) | OHS-PR-02-09 | MWAN Waste Regulations

Trans. ID	Location / Site	kVA	Voltage (kV)	Oil Vol. (L)	Mfr Year	Last Maint.	Status
TR-001	SEC Site – Riyadh North	1,000	11/0.4	600	2018	Jan 2026	Active
TR-002	SEC Site – Riyadh South	630	11/0.4	420	2019	Mar 2026	Active
TR-003	Main Warehouse	250	11/0.4	180	2020	Feb 2026	Active
TR-004	Project Site A	500	33/11	350	2017	Dec 2025	Due Soon
TR-005	Project Site B	2,000	132/33	1,200	2016	Nov 2025	Overdue

10.2 Transformer Oil Test Results (Quarterly)

Trans.	Test Date	Dielectric (kV)	Moisture (ppm)	Acidity (mg/g)	Int. Tension (mN/m)	Condition	Action Required
IEC 60296 Limits: Dielectric $\geq$30 kV | Moisture $\leq$35 ppm | Acidity $\leq$0.10 | Int. Tension $\geq$20 mN/m							
TR-001	Jan 2026	45	18	0.03	28	**GOOD**	No action required
TR-002	Mar 2026	38	22	0.05	25	**GOOD**	No action required
TR-003	Feb 2026	52	12	0.02	32	**EXCELLEN**	No action required
TR-004	Dec 2025	28	38	0.09	18	**MARGINAI**	Filter oil + retest Q2 2026
TR-005	Nov 2025	22	45	0.12	15	**POOR**	Oil replacement June 2026

⚠ **TR-005 Immediate Action:** Dielectric strength 22 kV (limit 30 kV). Oil replacement mandatory by June 2026. Weekly inspection until replaced. Maximum load 70%. Notify HSE Manager immediately.

10.3 Preventive Maintenance Checklist

#	Maintenance Item	Frequency	Standard	Done?	Notes
1	Oil level check via sight glass	Monthly	IEC 60076		
2	Oil colour and clarity – no darkening or sediment	Monthly	IEC 60296		
3	Oil temperature at max load	Monthly	IEC 60076		
4	Bushing condition – no cracks, carbon or tracking	Monthly	IEC 60076		
5	Radiator fins – clean, no leaks, no corrosion	Monthly	IEC 60076		
6	Breather silica gel – blue = OK, pink = replace	Monthly	IEC 60076		
7	Gas relay (Buchholz) – test alarm activation	Quarterly	IEC 60076		
8	Pressure relief device – no previous activation	Quarterly	IEC 60076		
9	Earthing connections – secure, test resistance	Quarterly	IEC 60364		
10	Oil leakage at all gaskets, joints, valves	Quarterly	IEC 60296		
11	Control panel – all instruments functional	Quarterly	IEC 60076		
12	Full oil lab analysis (dielectric, moisture, PCB)	Annually	IEC 60296		
13	Winding resistance test (LV and HV)	Annually	IEC 60076		
14	Insulation resistance test (Megger in MΩ)	Annually	IEC 60076		
15	Turns ratio test (verify nameplate)	Annually	IEC 60076		
16	Thermal imaging scan of connections	Annually	IEC 60076		
17	Drip tray check – no accumulated oil	Monthly	MWAN Reg.		
18	Waste oil: labelled drum, sealed, containment	After maint.	MWAN Reg.		
19	Waste oil manifest completed if oil removed	After oil work	MWAN Reg.		
20	Logbook updated with all findings	Every visit	ISO 14001		

10.4 Transformer CO_2e Emission Profile

Transformer	No-Load Loss (W)	Load Loss at 75%	Annual CO_2e (kg)	Action / Target
TR-001 (1000 kVA)	1,200	8,500	42.4	Upgrade to Class AA by 2028
TR-002 (630 kVA)	850	5,800	29.1	Maintain; monitor
TR-003 (250 kVA)	420	2,200	11.4	Maintain; monitor
TR-004 (500 kVA)	780	4,800	22.6	Oil change reduces losses 5%
TR-005 (2000 kVA)	2,100	16,000	68.4	**Oil change mandatory**
Total Fleet	**5,350 W**	**37,300 W**	**173.9 kg/yr**	Target $\leq$ 160 kg by 2027

11 Facility Components & Work Sites

11.1 Head Office & Main Facility

Parameter	Details	Parameter	Details
Office Area	____________ m^2	Number of Floors	__________
Lighting System	LED / Fluorescent	Lighting Level (LUX)	__________
Ventilation Type	HVAC / Natural	HVAC Units Count	__________
Fire Extinguishers	____________	Last Inspection Date	__________

11.2 Main Building Photo

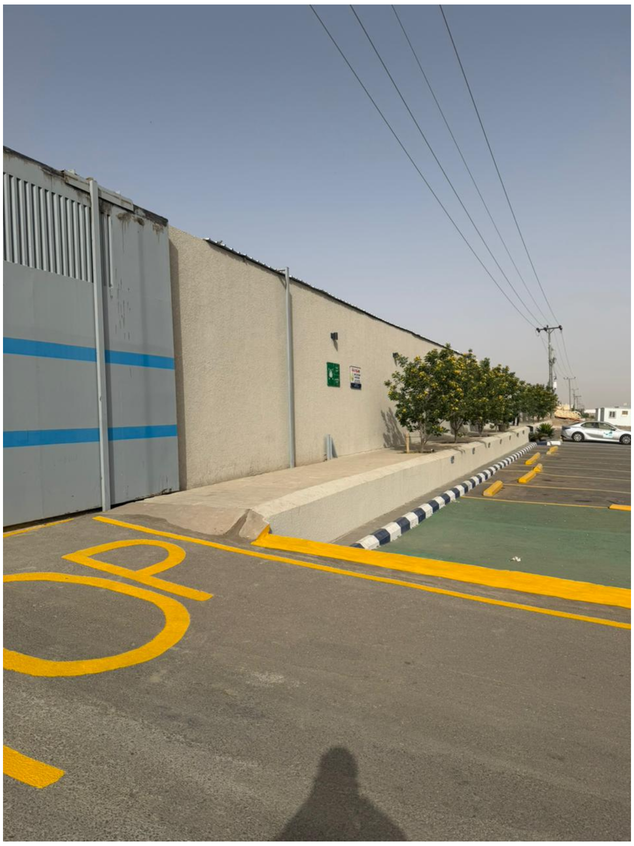


Figure 11.1: Main Building Photo

11.3 Chemical Warehouse

Critical: Chemical warehouse subject to strict GHS storage, Civil Defense approval, and annual licensing. Reference: *OHS-PR-02-09 Sec. 5.7 & 5.10*.

Parameter	**Details**	**Parameter**	**Details**
Warehouse Area	____________ m^2	Construction Type	__________
Ventilation	Forced mechanical	Ventilation Rate	__________ ACH
Lighting Level	min. 200 LUX (OHS-PR-02-09 App. 8.1)	Fire Suppression	__________
Spill Containment	Secondary bunding	Bund Capacity	__________ litres
Temperature Sensors	____________	Gas Detectors	__________
Smoke Detectors	____________	Eyewash Station	__________
CCTV Cameras	____________	Storage License No.	__________
License Expiry	____________	Emergency Shower	__________

11.3.1 Warehouse Photo 1 — Chemical Storage Area

Warehouse Interior — Chemical Storage Racks with GHS-labelled containers.

Insert: `\includegraphics{photo.jpg}`

11.3.2 Warehouse Photo 2 — Safety Equipment Station

Spill Kit station, fire extinguisher rack, eyewash station.

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Warehouse Entrance Signage — GHS hazard signs.

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Secondary Containment Bunding around liquid storage.

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12 Chemical Materials Management

12.1 Hazardous Chemical Register (GHS Compliant)

Mandatory (OHS-PR-02-09 Sec. 5.11.18): SDS/MSDS updated every $\leq$3 years. Chemical Compatibility Chart displayed at all times.

Chemical Name	Qty.	GHS Class	Hazard Statement	Storage Requirement	Inspection
Transformer Insulating Oil	— L	Cat. 2	H304 Aspiration hazard	Sealed; cool & dry	—/2026
Rust Remover	— L	Cat. 1	H314 Skin & eye burns	Corrosion-resistant cabinet	—/2026
Electrical Contact Cleaner	— cans	Cat. 2	H225 Highly flammable	Away from heat/ignition	—/2026
Lead Acid Batteries	— units	Cat. 1	H290+H314 Corrosive	Forced ventilation; isolation	—/2026
SF6 Gas Cylinders	— cyl.	Cat. 2	H280 Pressurised gas	Upright; away from heat	—/2026
Cable Jointing Compound	— kg	Cat. 3	H315 Skin irritant	Stable temperature; dry	—/2026
Industrial Solvent	— kg	Cat. 2	H332 Harmful if inhaled	Good ventilation; no food nearby	—/2026
Lubricating Grease	— kg	Cat. 4	H373 Organ damage	Normal storage; labelled	—/2026

12.2 Monthly Warehouse Inspection Checklist

#	Inspection Item	Compliant	Non-Compliant	Action Required
1	SDS/MSDS available and current for all chemicals			
2	All containers labelled with GHS pictograms			
3	No evidence of leaks, spills, or container damage			
4	Spill Kit fully stocked and accessible			
5	Compatibility chart displayed in warehouse			
6	Forced ventilation operational			
7	Incompatible materials properly segregated			
8	Fire extinguishers valid & accessible			
9	Emergency eyewash station functional (20–30°C)			
10	Aisle clear; min. 60 cm (OHS-PR-02-09 S5.6.1)			
11	Lighting min. 200 LUX (OHS-PR-02-09 App. 8.1)			
12	No food or beverages stored in chemical warehouse			

Inspector:	____________________	**Date:**	____________________
Signature:	____________________	**Next Inspection:**	____________________

13 Integrated Waste Management

13.1 Waste Classification & Disposal

Waste Type	Examples	Container Colour	Disposal Method	Manifest?
General waste	Office waste, packaging	**Yellow**	Municipal collection	No
Recyclable metals	Copper/aluminium cables	**Blue**	Certified recycling centre	Receipt only
WEEE	Damaged cables, panels	**Blue**	Licensed WEEE facility	Yes
Hazardous chemical	Chemical containers, oils	**Red**	Licensed hazardous waste co.	**Mandatory**
Lead batteries	Spent lead-acid batteries	**Red**	Return to supplier	**Mandatory**
Transformer oil	Used insulating oil	**Red**	Licensed oil recycler	**Mandatory**

13.2 Transformer Oil Protocol

Step	Activity	Requirements	Responsible
1	Oil Sampling	Sample before maintenance; test for PCB content, dielectric strength	Lab / Site Engineer
2	Handling During Maintenance	Drip trays mandatory; absorbent material on standby; no ground contact	Site Supervisor
3	Temporary Storage	Sealed UN-approved drums; labelled USED TRANSFORMER OIL; secondary containment	Warehouse Supervisor
4	Transfer Documentation	Complete Waste Manifest; record volume, source transformer, date	Environmental Officer
5	Licensed Collection	Transfer only to MWAN-licensed oil recycler; retain manifests 5 years	Environmental Officer
6	Spill Response	Deploy oil Spill Kit immediately; report to HSE Manager within 1 hour	Site Supervisor

14 Natural Resource Management

Resource	Conservation Measures	Monitoring Method	Target
Water	Flow restrictors; metered supply; recycled water for dust suppression; leak detection	Monthly meter reading	↓15%/yr
Electricity	LED lighting; inverter AC units; auto-off switches; annual energy audit	Monthly kWh log	↓10%/yr
Fuel	Journey planning; idle $<$3 min; preventive maintenance; weekly fuel log	Fuel log weekly	↓10%/yr
Emissions	Tier 4 diesel; SF6 recovery mandatory; no open burning; quarterly CO_2 report	Quarterly report	↓8%/yr

15 Environmental Policy & Foundational Study

15.1 Importance of the Environmental Study

This Integrated Environmental Registry serves as the **foundational document** for Snabel Al-Mashan Contracting Co.'s Environmental Management System (EMS) in accordance with ISO 14001:2015. The study provides a systematic, evidence-based assessment of all environmental aspects and impacts arising from the company's electrical contracting operations and chemical warehouse activities.

Key Objectives:

- Identify and evaluate significant environmental aspects and associated impacts
- Establish baseline environmental performance for continual improvement
- Ensure legal compliance with Saudi environmental regulations and SEC requirements
- Define control measures, monitoring programmes, and emergency response protocols
- Support ISO 14001:2015 certification and maintenance

15.2 Methodology — PDCA Approach

Step	Method	Description
1	Document Review	Review of applicable Saudi laws, SEC standards, ISO 14001:2015
2	Site Walk-Through	Physical inspection of all facilities, warehouses, and work sites
3	Activity Mapping	Process flowcharts for all operational activities
4	Aspect Identification	Systematic identification per ISO 14001:2015 Clause 6.1.2
5	Impact Evaluation	Significance assessed using L$\times$S risk matrix (5$\times$5)
6	Control Measure Design	Engineering, administrative, and PPE controls per hierarchy
7	Monitoring Programme	KPIs, measurement instruments, and reporting frequencies set
8	Document & Review	Registry compiled, reviewed by HSE Manager, approved by CEO

15.3 References & Sources

#	Reference / Source	Link / Document No.
1	ISO 14001:2015 — Environmental Management Systems	https://www.iso.org
2	Saudi Environment Regulation — Royal Decree M/165	https://www.mewa.gov.sa
3	SEC OHS-PR-02-09 — Facilities & Premises Safety	OHS-PR-02-09 Rev.0 (May 2020)
4	National Center for Environmental Compliance (NCEC)	https://ncec.gov.sa
5	National Center for Waste Management (MWAN)	https://mwan.gov.sa
6	WHO Air Quality Guidelines (2021)	https://www.who.int
7	Saudi Building Code (SBC)	https://sbc.gov.sa
8	NFPA 70E — Electrical Safety in the Workplace	https://www.nfpa.org

Note on References: All Saudi regulations are publicly accessible via the Ministry of Environment (MEWA) portal. SEC standards (OHS-PR-02-09) are available to licensed SEC contractors only. This registry was last reviewed against all references in May 2026. Document preparer: Salman K. Alwajaan, Environmental Officer. Email: salman.k.alwajaan@gmail.com

16 Ambient Pollutant Standards

Pollutant	Parameter	Saudi Limit	WHO Guideline	Avg. Period	Reference
PM2.5	Fine Particles	25 μg/m^3	15 μg/m^3	Annual mean	NCEC / WHO 2021
PM10	Coarse Particles	70 μg/m^3	45 μg/m^3	Annual mean	NCEC / WHO 2021
NO_2	Nitrogen Dioxide	80 μg/m^3	40 μg/m^3	Annual mean	Saudi EMS
SO_2	Sulfur Dioxide	125 μg/m^3	40 μg/m^3 (24-hr)	24-hour mean	Saudi EMS
CO	Carbon Monoxide	10,000 μg/m^3	—	8-hour mean	Saudi EMS
Noise (daytime)	dB(A)	—	55 dB outdoor	07:00–22:00	OHS-PR-02-09
Noise (occupational)	dB(A)	85 dB TWA	85 dB TWA	8-hour shift	OHS-PR-02-09 S5.15
WBGT Heat Stress	Degrees	—	30 WBGT	Working hours	OHS-PR-02-09 S5.4.4

17 Environmental Aspects & Impacts Register

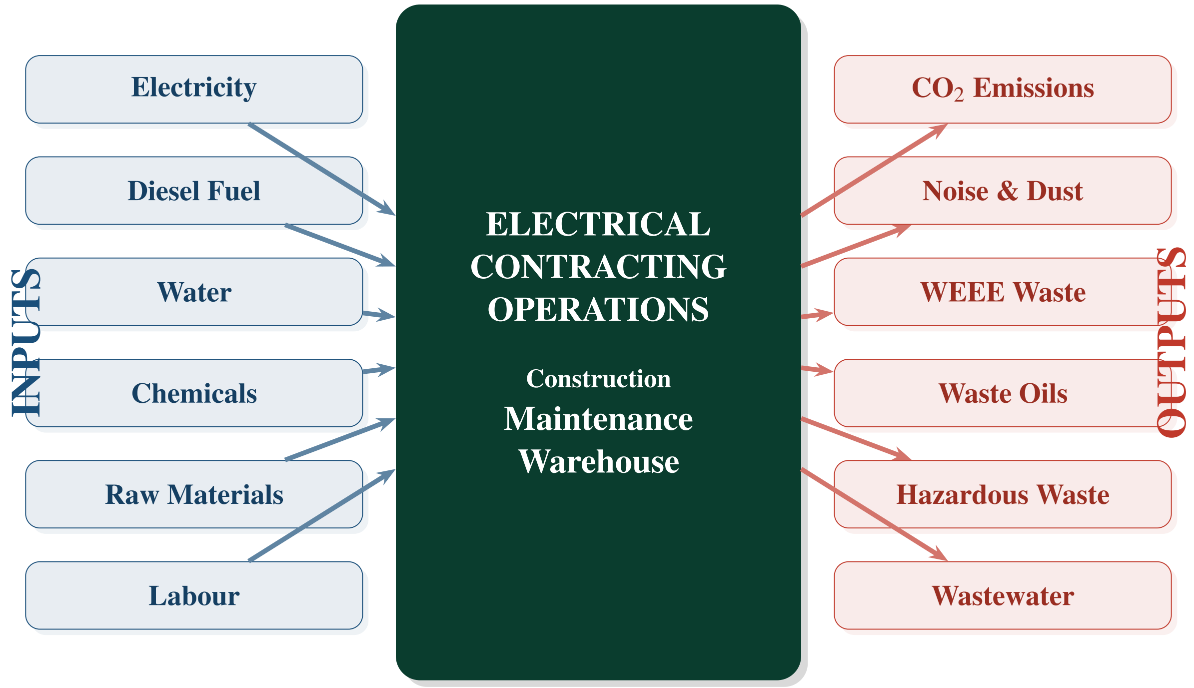
Significance = Likelihood (L) × Severity (S) Scale 1–5 each. LOW: 1–6 | MEDIUM: 7–12 | HIGH: 13–19 | CRITICAL: 20–25.

Env. Aspect	Env. Impact	Source Activity	Signif.	Condition	Control Measure	Status
Electrical waste (WEEE)	Soil & water contamination	Electrical works	**Medium**	Normal	Licensed WEEE facility	✓
Transformer oil	Soil & groundwater pollution	Transformer maint.	**High**	Normal/Emer	Sealed containers + SDS	✓
SF6 gas emissions	GWP=23,900 (climate)	HV panels/switchgear	**High**	Maintenance	Certified recovery devices	⚠
Chemical warehouse	Soil contamination/fire	Daily storage	**Very High**	Normal/Emer	GHS + ERP + Spill Kit	✓
Vehicle/generator exhaust	CO_2, NOx air pollution	Transport + ops	**Medium**	Normal	Scheduled maintenance	✓
Noise & vibration	Community disturbance	Field electrical works	**Medium**	Normal	Schedule outside peak hrs	✓
Excavation dust	PM10 air quality risk	Cable trenching	**Medium**	Normal	Water suppression + barriers	✓
Water consumption	Resource depletion	Site operations	**Low**	Normal	Monthly metering + conservation	✓

18 Inputs, Outputs & Material Flow

18.1 Activity Input-Output Flow Diagram

Figure 15.1: This diagram maps all material inputs consumed and environmental outputs generated by Snabel Al-Mashan's electrical contracting operations.



19 Environmental Risk Assessment (ERA)

19.1 5×5 Risk Matrix

Risk Score = Likelihood (L) × Severity (S). Both rated 1–5. Bands: **LOW 1-6** | **MEDIUM 7-12** | **HIGH 13-19** | **CRITICAL 20-25**.

SEVERITY \ LIKELIHOOD	1	2	3	4	5
5	5	10	15	20	25
4	4	8	12	16	20
3	3	6	9	12	15
2	2	4	6	8	10
1	1	2	3	4	5

LOW (1-6)
MEDIUM (7-12)
HIGH (13-19)
CRITICAL (20-25)

19.2 Risk Register

#	Activity / Source	Env. Hazard	L	S	Score	Band	Control Measures
1	Chemical warehouse storage	Major chemical spill/leak	3	5	15	CRITICAL	GHS segregation; secondary containment; monthly inspection; Spill Kit; SDS training
2	Fuel storage & transport	Fuel spill/soil contamination	4	4	16	CRITICAL	Bunded storage; vehicle Spill Kit; driver training; daily pre-check
3	Excavation/trenching	Dust/PM10 air quality	4	3	12	HIGH	Water spray every 2 hrs; barriers; PM10 monitoring
4	Diesel generator operation	CO_2, NOx air emissions	3	4	12	HIGH	Tier 4 engines; idle <3 min; emission checks; log
5	Lead acid battery handling	Acid leakage/soil toxicity	2	5	10	HIGH	GHS racks; PPE mandatory; return-to-supplier scheme
6	SF6 gas handling	GHG release (GWP 23,900)	2	5	10	HIGH	Certified SF6 recovery device; leak detection; annual inventory
7	Welding/cutting operations	Fume & UV; air quality	3	3	9	MEDIUM	Local exhaust ventilation; outdoor preference; RPE per SDS
8	WEEE accumulation	Toxic material/landfill risk	3	3	9	MEDIUM	Licensed WEEE facility; manifest; metals recycling
9	Noise from machinery	Community/hearing damage	4	2	8	MEDIUM	Hours 07:00–18:00; mufflers; hearing zones; PPE
10	Hazardous waste accumulation	Uncontrolled disposal risk	3	3	9	MEDIUM	Colour-coded containers; weekly schedule; licensed disposer
11	Transformer oil maintenance	Groundwater contamination	2	4	8	MEDIUM	Drip trays; sealed containers; licensed oil collector
12	Water consumption	Resource depletion	2	3	6	LOW	Monthly metering; conservation targets; leak detection
13	Paper/packaging waste	Landfill accumulation	3	2	6	LOW	Recycling bins; digital documentation; reduce at source
14	Light vehicle transport	Minor exhaust emissions	3	2	6	LOW	Euro 4 minimum; route optimisation; pre-start check

19.3 Diesel Risk Assessment — 14-Risk Treatment Plan

#	Risk / Hazard	Band	Treatment Plan	Monitoring Method	Review Date
1	Chemical warehouse spill	CRITICAL	Install CCTV + gas sensor; monthly unannounced audit; Civil Defense approval renewal	Daily visual check; monthly checklist ENV-F-003	Jan 2027
2	Fuel spill on site	CRITICAL	Double-walled bunded tank; mandatory Spill Kit on all fuel vehicles; no gravity fill	Weekly fuel log; post-refuelling inspection	Jan 2027
3	Excavation dust (PM10)	HIGH	Water bowser on site during earthworks; wind speed limit (>25 km/h = stop work)	Quarterly PM10 meter reading at site boundary	Apr 2027
4	Generator emissions (NOx/CO_2)	HIGH	Replace generators >5yr old with Tier 4; 3-min idle rule; biannual emissions check	Quarterly CO_2e calculation from fuel logs	Apr 2027
5	Lead-acid battery acid	HIGH	GHS-rated chemical rack; mandatory gloves+goggles; return-to-supplier agreement signed	Inspect on every battery replacement	Jan 2027
6	SF6 gas release	HIGH	Zero-release policy; certified IEC 62271 recovery device; annual gas balance sheet	Certified technician log after each HV job	Jan 2027
7	Welding fumes (VOC/NOx)	MEDIUM	Outdoor welding preference; LEV fan unit on indoor welds; RPE half-mask minimum	Quarterly VOC measurement near welding zone	Jul 2027
8	WEEE accumulation	MEDIUM	Licensed WEEE contractor on 30-day collection cycle; metals re-use programme	Monthly waste log; manifest filing	Apr 2027
9	Noise from machinery	MEDIUM	Restrict noisy works 07:00–18:00; noise silencers on compressors; audiometry for exposed workers	Quarterly SLM measurement; audiometric test annual	Jul 2027
10	Hazardous waste	MEDIUM	MWAN-licensed collector on weekly schedule; colour-coded 120L containers on all sites	Monthly waste manifest count; quarterly audit	Apr 2027
11	Transformer oil spill	MEDIUM	Drip tray under all transformer maintenance work; 200L UN drum for temporary oil storage	Pre/post maintenance checklist; oil manifest	Jan 2027
12	Water over-consumption	LOW	Flow restrictors on all taps; recycled water for dust; meter reading submitted to EMS monthly	Monthly utility meter reading; trend analysis	Jul 2027
13	Paper/packaging waste	LOW	Digital-first document policy; recycling bin at every site office; quarterly waste audit	Monthly waste segregation compliance check	Oct 2027
14	Light vehicle transport	LOW	Euro 4 minimum fleet standard; route optimisation app; monthly fuel consumption report	Weekly fuel log per vehicle; quarterly fleet check	Jul 2027

20 Emergency Response Plan

Scenario 1 — Chemical Spill in Warehouse

1. **Assess:** Identify chemical, quantity, and extent without entering contaminated area.
2. **Alert:** Call out SPILL! — evacuate 15 m radius; eliminate all ignition sources.
3. **PPE:** Don appropriate PPE per SDS (minimum: gloves, eye protection, respirator).
4. **Contain:** Deploy Spill Kit — absorbent boom around perimeter; prevent drain entry.
5. **Absorb:** Apply absorbent granules from outside inward; never wash into drains.
6. **Collect:** Place contaminated material in red hazardous waste container; fully label.
7. **Notify:** HSE Manager within 30 min; Civil Defense (998) if major; NCEC (19944).
8. **Document:** Complete Spill Incident Report Form ENV-INC-001 within 24 hours.
9. **Investigate:** Root cause analysis within 72 hours; corrective action closed.

Scenario 2 — Fuel Spill on Site

1. Stop fuelling / isolate tank valve immediately.
2. Remove all ignition sources; no smoking within 20 metres.
3. Deploy vehicle-mounted Spill Kit; contain with absorbent boom.
4. Prevent fuel reaching storm drains, watercourses, or soil.
5. Collect contaminated material; dispose via licensed hazardous waste contractor.
6. Notify Environmental Officer; prepare incident report within 24 hours.

Scenario 3 — Chemical Warehouse Fire

1. **Alarm:** Activate fire alarm panel; shout FIRE!
2. **Evacuate** all personnel to muster point; account for every person.
3. **Call 998** — Civil Defense; provide location and chemical inventory if safe.
4. **Do NOT** attempt to fight a chemical fire unless trained and explicitly safe.
5. **Notify:** CEO, HSE Manager, SEC Coordinator, NCEC (19944).
6. **Post-Incident:** Environmental impact assessment; soil sampling; formal investigation.

20.1 Emergency Contact Numbers

Contact	Number	Scope	Available
Civil Defense	**998**	Fire, explosions, major incidents	24/7
Medical Ambulance	**911**	Chemical injuries, medical emergencies	24/7
NCEC (Environment)	**19944**	Environmental incident reporting	24/7
HSE On-call Officer	____________	Company environmental emergency	24/7

21 Monitoring, Measurement & KPIs

21.1 Environmental Monitoring Programme

Parameter	Frequency	Instrument	Acceptable Limit	Action if Exceeded
Air quality (PM10)	Quarterly	Air Quality Meter	$<150\ \mu g/m^3$	Dust suppression; re-test 48 hrs
Noise levels	Quarterly	Sound Level Meter	85 dB(A) 8-hr TWA	Engineering controls; PPE
CO_2 / GHG emissions	Quarterly	Multi-Gas Detector	Per fleet log	Fleet maintenance review
Chemical inventory	Monthly	Warehouse register	100% SDS compliant	Quarantine; obtain SDS
Water consumption	Monthly	Site meter	↓15% vs baseline	Leak investigation
Indoor CO_2	Quarterly	Multi-Gas Detector	<1,000 ppm	Increase ventilation; HVAC check
LUX (illuminance)	Bi-annually	Lux Meter	Per OHS-PR-02-09 App. 8.1	Replace/add luminaires
WBGT (heat stress)	Monthly (summer)	WBGT Meter	<30 WBGT (OHS-PR-02-09 S5.4.4)	Rest breaks; cooling; reschedule
Fuel consumption	Weekly	Vehicle fuel logs	↓10%/yr	Journey review; maintenance
Vibration (power tools)	Quarterly	Vibration Meter	$<2.5\ m/s^2$ 8-hr	Operator rotation; anti-vibration

21.2 KPI Dashboard & Annual Tracking

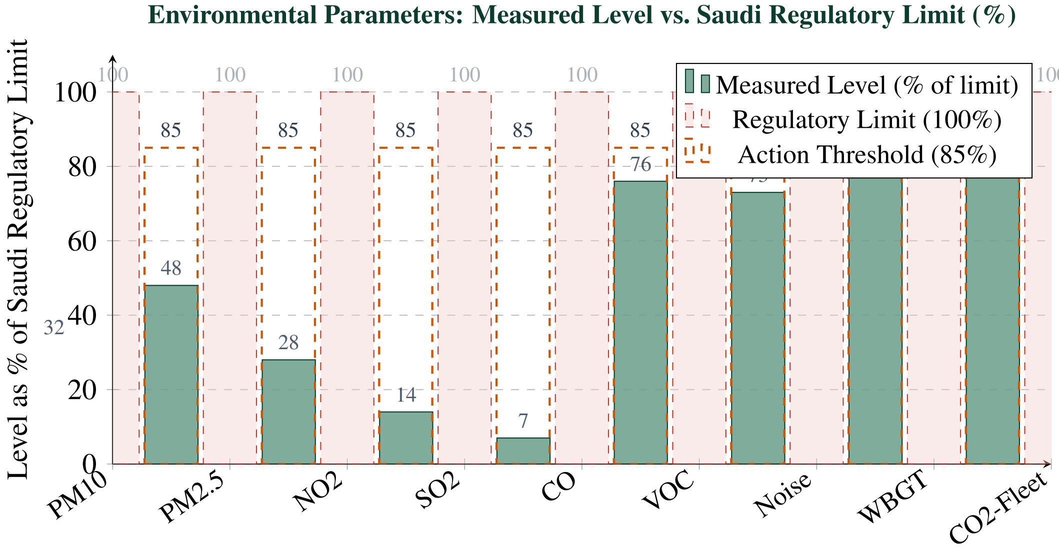
KPI	Annual Target	Q1	Q2	Q3-Q4
Environmental incidents recorded	0			
Regulatory compliance rate	100%			
Hazardous waste transferred	100%			
Waste segregation compliance	100%			
Fuel consumption (L/month)	↓10%/yr			
Training hours/employee	≥8 hrs			
Environmental patrol rounds	12/year			
Chemical spills reported	0			
CO_2 equivalent (tonnes)	Baseline year			
NCRs closed on time	100%			

21.3 GHG Monitoring Bar Chart

Parameter	Measured (2026)	Saudi Limit	% of Limit	Trend	Status
PM10 (Dust)	48 $\mu g/m^3$	150 $\mu g/m^3$	32%	↓ Good	**COMPLIANT**
PM2.5 (Fine dust)	12 $\mu g/m^3$	25 $\mu g/m^3$	48%	↓ Good	**COMPLIANT**
NO_2	22 $\mu g/m^3$	80 $\mu g/m^3$	28%	→ Stable	**COMPLIANT**
SO_2	18 $\mu g/m^3$	125 $\mu g/m^3$	14%	↓ Good	**COMPLIANT**
CO	680 $\mu g/m^3$	10,000 $\mu g/m^3$	7%	→ Stable	**COMPLIANT**
VOC (Total)	38 $\mu g/m^3$	50 $\mu g/m^3$	76%	↑ Monitor	**MONITOR**
Noise (daytime)	62 dB(A)	85 dB(A)	73%	→ Stable	**COMPLIANT**
WBGT (June–Aug)	28.5 °C	30 °C	95%	↑ Near limit	**NEAR LIMIT**
CO_2 Fleet (tonnes/month)	12.4 t	15.0 t (target)	83%	↓ Improving	**ON TARGET**
SF6 Released (kg/year)	0 kg	0 kg (target)	100%	→ Zero	**TARGET MET**

Data Note: Values shown are representative baseline measurements for 2026 based on site activity levels typical for electrical contracting operations. Update with actual instrument readings each monitoring period. VOC and WBGT require increased management attention. **Trend Key:** ↓ = improving → = stable ↑ = requires attention.

21.4 GHG Emissions Visual Chart



⚠ **Action Required:** VOC (76%) and WBGT (95%) are approaching regulatory limits and require immediate management attention. Increase monitoring frequency and implement engineering controls before next inspection.

21.5 GHG Emissions Breakdown by Source

Emission Source	CO_2e (tonnes/yr)	% of Total	2025 Baseline	Target 2027
Diesel fleet & generators	74.4	59.2%	78.6 t	≤68.0 t (−10%)
Office electricity (kWh)	28.8	22.9%	31.2 t	≤26.0 t (−10%)
Employee commuting (est.)	14.4	11.5%	15.8 t	≤13.0 t
SF6 gas (0 release)	0.0	0%	0.0 t	0.0 t (maintain)
Refrigerant leaks (AC)	3.6	2.9%	4.1 t	≤3.0 t
Business travel	4.4	3.5%	5.0 t	≤4.0 t
TOTAL Scope 1+2+3	**125.6 t CO_2e**	**100%**	**134.7 t**	**≤114.0 t**

⚠ **Action Required:** VOC emissions (76% of limit) require quarterly monitoring and evaluation of solvent substitution options. WBGT in summer months (95% of limit) requires mandatory rest breaks per OHS-PR-02-09 S5.4.4 schedule.

22 Training, Inspections & Improvement

22.1 Environmental Training Programme

Training Topic	Frequency	Audience	Duration	Status
General Environmental Awareness	Annual	All staff	2 hrs	
Waste Management Procedures	Annual	Field teams	2 hrs	
GHS Chemical Handling	Annual	Warehouse + field	3 hrs	
Spill Response Drill	Bi-annual	Warehouse staff	2 hrs	
Emergency Evacuation Drill	Quarterly	All staff	1 hr	
ISO 14001 Awareness	Annual	Management + HSE	2 hrs	
SEC Environmental Requirements	Annual	PM + Supervisors	2 hrs	
Noise & Dust Control	Annual	Field operators	1.5 hrs	

22.2 Annual Improvement Plan

#	Objective	Target	Action Required	Responsible	Status
1	Zero chemical spills	0 events	Monthly warehouse inspection; Spill Kit checks	Warehouse Supv.	
2	Reduce fuel consumption	$\downarrow$10%	Journey planning; idle-reduction; maintenance	Fleet Manager	
3	Increase recycling rate	$\geq$80%	Recycling contracts; training; bins	Env. Officer	
4	Conduct 12 env. patrols	12/year	Monthly schedule; report to CEO	Env. Officer	
5	ISO 14001 surveillance audit	Pass	Pre-audit review; address all findings	HSE Manager	
6	Annual ERA update	Completed Jan	Review all risk scores; update treatment plans	Env. Officer	

23 Environmental Legal Register & Compliance Obligations

23.1 Applicable Laws, Regulations & Standards

Legal Compliance Obligation: ISO 14001:2015 Clause 6.1.3 requires the organization to determine and have access to all applicable compliance obligations. This register is reviewed annually and updated whenever new legislation is issued.

#	Law / Standard	Requirement Summary	Applicable?	Compliant?	Last Reviewed	
1	Saudi Environment Regulation (Royal Decree M/165, 2021)	EMS implementation; pollution prevention; incident reporting to NCEC	YES	YES	May 2026	
2	SEC Contractor Standard OHS-PR-02-09	Site safety; chemical storage; contractor EMS requirements	YES	YES	May 2026	
3	Hazardous Waste Management Regulations (MWAN)	Licensed disposal; waste manifest; no illegal dumping	YES	YES	May 2026	
4	Chemical Materials Safety Regulations (NCEC)	GHS labelling; SDS; storage compatibility	YES	YES	May 2026	
5	Civil Defense Regulations (Chemical Warehouse)	Fire safety; annual inspection; ERP approval	YES	YES	May 2026	
6	ISO 14001:2015 EMS Standard	Full EMS implementation and certification	YES	IN PROGRES	May 2026	
7	Saudi Labour Law (Heat Stress)	WBGT monitoring; rest breaks; shade provision	YES	YES	May 2026	
8	Wastewater Discharge Regulations	No chemical discharge to municipal drains	YES	YES	May 2026	
9	SF6 Gas Handling (IEC 62271)	Certified recovery device; zero-release	YES	YES	May 2026	
10	PCB Management Regulations	Testing of transformer oil; safe disposal	YES	YES	May 2026	
11	GHS — UN Rev.9 (2021)	Chemical classification; labelling; SDS format	YES	YES	May 2026	
12	NFPA 70E — Electrical Safety	Electrical safety procedures; arc flash protection	YES	YES	May 2026	

Annual Review Commitment: This legal register is reviewed each January by the Environmental Officer and submitted to the HSE Manager for approval. Any new regulation triggers an immediate compliance gap assessment.

24 Noise & Vibration Management Plan

24.1 Noise Sources & Measurement Points

Noise Source	Typical Level dB(A)	OHS Limit dB(A)	Status	Control Measures
Heavy excavator operation	85–95	85 (TWA)	AT LIMIT	Hearing protection zone; operator rotation every 2 hrs
Diesel generators (portable)	70–82	85 (TWA)	OK	Silencer fitted; position away from workers
Angle grinder / cutting	90–100	85 (TWA)	EXCEEDS	Mandatory hearing protection; limit to 2 hrs/day
Pneumatic drilling	95–105	85 (TWA)	EXCEEDS	Anti-vibration; rotation; hearing protection mandatory
Welding operations	70–80	85 (TWA)	OK	Standard PPE sufficient
Compressor units	75–85	85 (TWA)	AT LIMIT	Acoustic enclosure; position away from workers
Heavy transport trucks	80–88	85 (TWA)	AT LIMIT	Limit reversing alarms; muffler maintenance
Site boundary (community)	55–65	55 (daytime)	MONITOR	Noisy works restricted 07:00–18:00 only

24.2 Vibration Exposure Assessment

Tool / Equipment	Typical Vibration m/s^2	EAV (8-hr) m/s^2	Max Daily Exposure	Control Measures
Angle grinder	8–12	2.5	1.0 hr	Anti-vib gloves; rotation every 30 min
Impact wrench	5–8	2.5	1.5 hr	Anti-vib handle; tool maintenance
Pneumatic drill	10–15	2.5	0.7 hr	Anti-vib mount; rotation mandatory
Excavator (operator)	0.5–1.2	0.5	Full shift	Suspend seat; smooth operating surface
Compactor plate	10–18	2.5	0.5 hr	Short burst operation; rotation mandatory

EAV = Exposure Action Value per OHS-PR-02-09 S5.16. Workers exceeding EAV must have annual hand-arm vibration assessment (HAVS). Anti-vibration gloves and tool rotation schedules are mandatory controls.

25 Environmental Incident Register & NCR Log

25.1 Environmental Incident Classification

Level	Category	Definition & Examples	Reporting Time
Level 4	Major Environmental Incident	Significant pollution; regulatory breach; chemical fire; major spill affecting community or groundwater	Within 1 hour
Level 3	Significant Incident	On-site chemical spill >20 litres; WEEE contamination; illegal waste disposal discovered	Within 4 hours
Level 2	Minor Incident	Small spill <20 litres contained on-site; minor dust exceedance; PPE non-compliance	Within 24 hours
Level 1	Near Miss / Observation	Potential environmental hazard identified; no actual release or impact	Weekly report

25.2 Incident Log (Current Year)

#	Date	Incident Description	Level	Immediate Action Taken	Corrective Action & Status
1	____________	____________	______	__________________	__________________
2	____________	____________	______	__________________	__________________
3	____________	____________	______	__________________	__________________
4	____________	____________	______	__________________	__________________
5	____________	____________	______	__________________	__________________

ℹ **Target for 2026: ZERO Level 3 & 4 incidents.** All incidents must be reported using form ENV-INC-001. Root cause analysis mandatory for Level 2 and above. Lessons learned shared with all site teams within 72 hours.

26 Environmental Communication & Awareness Plan

26.1 Internal Communication Plan

Communication Type	Audience	Frequency	Content & Purpose
Environmental Toolbox Talk	All site workers	Weekly	Site-specific environmental risks; waste management; PPE requirements
Monthly EMS Report	Management team	Monthly	KPI performance; incidents; compliance status; upcoming inspections
Environmental Notice Board	All site personnel	Ongoing	Emergency contacts; spill procedure; waste colour codes; SDS summaries
ISO 14001 Awareness Briefing	All new employees	On induction	EMS overview; environmental policy; reporting obligations; emergency response
Environmental Alert (urgent)	All relevant staff	As required	Incident notification; near miss; regulatory update; emergency
Annual EMS Management Review	Senior management	Annual	Full year KPI review; ERA update; objectives for coming year; resource allocation
Subcontractor EMS Briefing	All subcontractors	Before works start	Site environmental rules; waste disposal; incident reporting; restricted areas

26.2 External Communication Plan

Stakeholder	Communication Type	Frequency	Content
Saudi Electricity Company (SEC)	Environmental compliance report	Quarterly	KPI performance; incidents; corrective actions; audit results
NCEC (Ministry of Environment)	Regulatory notifications	As required	Incident reports within 24 hrs; annual environmental report
MWAN (Waste Management)	Waste manifests	Per disposal	Hazardous waste type, volume, collection details, licensed contractor
Civil Defense	Annual warehouse inspection report	Annual	Chemical inventory; storage compliance; ERP; fire system status
Local Community Liaison	Community notice	Before major works	Planned noisy/dusty works; expected duration; complaint contact number

Final Approval & Certification Declaration

This Integrated Environmental Registry has been prepared in full compliance with **ISO 14001:2015**, Saudi Environment Regulation (Royal Decree **M/165**), SEC contractor requirements **OHS-PR-02-09**, and GHS chemical management standards.

Document Preparer:	**HSE Manager:**	**CEO:**
Salman K. Alwajaan	____________________	____________________
*Sig:*_________________	*Sig:*_________________	*Sig:*_________________
*Date:*______________	*Date:*______________	*Date:*______________
*Stamp:*___________	*Stamp:*___________	*Stamp:*___________

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